

Saaras Pakanati

Cincinnati, Ohio, USA

email: pakanass@mail.uc.edu
github: [SaarasPakanati](#)

Education	University of Cincinnati, Cincinnati, Ohio B.S. in Mechanical Engineering Minor in Mathematics GPA 3.95/4.00	August 2021 - Present
Research Interests	Computational Fluid Dynamics, Multiphase Flows, Machine Learning, Data Driven Modeling	
Research Experience	Undergraduate Research Assistant UC Lab for Interfacial Dynamics, advised by Dr. Kishan Bellur Dept. of Mechanical & Materials Engineering, University of Cincinnati	August 2024 - Present
	• Numerical & Experimental study of multiphase interfaces and associated phenomenon. • Modeled multiscale cryogenic fuel evaporation using custom UDFs in ANSYS Fluent. • Designed & built an interferometry setup to study instabilities in liquid thin films. • Assisting in the study of the effect of temperature-sensitive surfactants on interfacial dynamics.	
Peer-Reviewed Publications	1. A. Yasin*, S. Pakanati* , K. Bellur, <i>A multiscale CFD model of evaporating Hydrogen menisci: Incorporating subgrid thin-film dynamics and in situ accommodation coefficients</i> , Fuels, 2026 *equal contribution https://doi.org/10.3390/fuels7010003	
Conference Presentations	2. A. Sarchami, S. Pakanati , T. Enam, A. Yasin, K. Bellur, <i>Stability and Contact Line Dynamics of Evaporating Thin Liquid Films</i> , 11 th ASTFE Thermal and Fluids Engineering Conference, Tempe, Arizona, March 9 – 12, 2026 1. A. Sarchami, S. Pakanati , K. Bellur, <i>Dynamic Optical Measurement of Evaporating Liquid Films</i> , ASME International Mechanical Engineering Congress & Exposition, Memphis, Tennessee, November 16 – 20, 2025 (Poster)	
Honors, Awards, and Fellowships	Merchant, M. Eugene Scholarship Awarded by the College of Engineering and Applied Science Armstrong Discovery Fellowship Awarded by the Armstrong Institute for Space, Technology, and Research Undergraduate Research co-op Fellowship (2×) Awarded by the College of Engineering and Applied Science CEAS Annual Scholar - Undergraduate Research Awarded by the College of Engineering and Applied Science UC Mathematics Department Calculus Contest - Runner's Up Awarded by Cengage Publishing & UC Department of Mathematical Sciences CEAS International Outreach Scholarship Awarded by the College of Engineering and Applied Science University of Cincinnati Global Scholarship Awarded by the University of Cincinnati	2026 2025-2026 2024-2025 2025 2022 2021 2021
Teaching Experience	Grader Dept. of Mechanical & Materials Engineering, University of Cincinnati	August 2025 - December 2026
	• Course(s): MECH 2010 (Thermodynamics) • Managed assignments and proctored exams for 150+ students.	

Supplemental Review Session Instructor
University of Cincinnati Learning Commons

January 2023 - May 2025

- Led in-class review sessions for Calculus I and II.
- Tutored students in 15 different Engineering, Mathematics, Physics, and Chemistry courses.

Industry Experience

New Product Development Mechanical Engineering co-op
MSA Safety, Cranberry Township, Pennsylvania

January 2024 - April 2024

- Executed Finite Element Analysis in Ansys to validate design decisions for safety devices.
- Optimized prototype testing of Technology Readiness Level (TRL) 1 gas detection technologies by developing and applying Design of Experiments (DOE) methods using Minitab.
- Collaborated with Electrical and Software teams for cross-functional design.

Engineering co-op

Richards Industrials, Cincinnati, Ohio

August 2022 - December 2022 & May 2023 - August 2023

- Tested critical load-bearing components using destructive and non-destructive methods.
- Created technical drawings for shop floor fixtures using SolidWorks, collaborating with machinists.
- Developed tools and reports for engineering calculations, process optimization, and quality analysis.

Software Intern

Visual Collaboration Technologies, Inc., Bengaluru, Karnataka, India

June 2022 - July 2022

- Created a visualization tool for a CFD study using the dash framework in Python.
- Collaborated in an Agile development framework to code, following PEP8 standards.

Skills

Simulation: Ansys (Fluent, Static Structural, LS Dyna), Siemens simulation suite (Siemens NX, StarCCM+), OpenFOAM, ParaView, SolidWorks, LAMMPS

Programming: Python, MATLAB, C, C++, Bash, R Language, Minitab, LabView, $\text{\LaTeX}$

Experimental: Optical interferometry, image processing and analysis, flow control, Semiconductor manufacturing, Design of Experiments

Certifications & Licenses

OASiS Foundations Program, *University of Cincinnati* – 2025 | [link to certificate](#)

MATLAB Fundamentals, *Mathworks* – 2023 | [link to certificate](#)

IBM Data Science Practitioner, *IBM* – 2022 | [link to certificate](#)

Six Sigma Yellow Belt, *The Council for Six Sigma Certification* – 2022 | Credential ID: h7L0bACNv5

NX – CAD Design Associate, *Siemens* – 2022 | Credential ID: 6763-164-884-2231

Extra-Curricular

President

Undergraduate Research Society, University of Cincinnati

September 2021 - Present

- President (2025-26), VP (2024-25) and, Events Chair (2022-23 & 2023-24)
- Organized meetings with executive team; managing presentations and logistics for regular outreach.
- Gave multiple presentations on undergraduate research at university classes, general meetings, and academic organizations.

Volunteer Tutor

Bearcat Buddies, Cincinnati, Ohio

October 2021 - December 2023

- Tutored students in the Cincinnati Public School system and assisted in the improvement of the content covered with the local in-charge.